Symbols

A	Area
a	Thermal diffusivity
a	Throttling factor (extrusion)
a_w	Water activity
B	Time of heating (canning)
Bi	Biot number
b	Permeability
b	Slope of sorption isotherm
C_d	Drag coefficient (fluid dynamics)
c	Concentration
c	Internal seam length (canning)
c	Specific heat capacity
c_p	Specific heat at constant pressure
D	Diameter (pipe, vessel)
D	Dilution rate (fermentation)
D	Decimal reduction time
D	Diffusion coefficient
d	Diameter (sphere, size of sieve aperture)
d	Differential operator
E	Electrical field strength
E	Energy (size reduction, radio frequency heating)
F	Feed flow rate (sorting, fermentation)
F	F-value (canning)
F	Shape factors (extruders)
Fr	Froude number
f	Slope of heat penetration curve (canning)
f	Frequency (microwaves)
G	Geometric constants (extruders)
G	Air mass flowrate (dehydration)
g	Acceleration due to gravity (9.81 m s^{-2})
g	Retort temperature minus product temperature (canning)
H	Humidity

h	Heat transfer coefficient
h_c	Convective heat transfer coefficient
h_s	Surface heat transfer coefficient
I	Light intensity
I	Electrical current
I_h	Retort temperature minus product temperature (canning)
J	Flux (membrane concentration)
j	Heating/cooling factor (canning)
K	Mass transfer coefficient (dehydration, membrane concentration)
K	Constant
K_k	Kick's constant (size reduction)
K_R	Rittinger's constant (size reduction)
K_s	Substrate utilisation constant (fermentation)
k	Thermal conductivity
L	Length
L	Equivalent thickness of filter cake
l	Come-up time (canning)
M	Moisture content, dry-weight basis
M	Molar concentration
m	Mass
m	Mass flow rate
m	Moisture content (wet-weight basis)
N	Speed
N	Rate of diffusion
Nu	Nusselt number
n	Number
P	Pressure
P	Product Row rate (sorting)
P	Power
P	Productivity (fermentation)
Po	Power number (mixing)
Po	Vapour pressure of pure water
Q	Rate of heat transfer
Q	Volumetric flowrate
q_p	Specific rate of product formation (fermentation)
R	Universal gas constant
R	Reject flowrate (sorting)
R	Resistance to flow through a filter
R	Fraction of reflected light (packaging)
R	Electrical resistance
Re	Reynolds number
r	Radius
r	Specific resistance to flow through a filter
S	Substrate concentration (fermentation)
s	Compressibility of filter cake
T	Absolute temperature
T	Fractional transmission of light (packaging)
t	Time
t	Metal thickness (canning)
U	Overall heat transfer coefficient
U	Thermal death time at retort temperature (canning)
V	Volume
V	Voltage

V_c	Fractional volume of filter cake
v	Velocity
v_e	Air velocity needed to convey particles
v_f	Air velocity needed for fluidisation
W	Work index (size reduction)
x	Thickness, depth
x	Direction of heat flow
$\underline{x}$	Mass fraction
$\bar{x}$	Average
y	Cover hook length (canning)
Y	Yield or yield factor (fermentation)
z	Height
z	z-value (canning)
α	Absorbance, absorptivity
β	Coefficient of thermal expansion
Δ	Difference, change
δ	Half dimension
$\tan\delta$	Loss tangent (microwaves)
ϵ	Porosity
ϵ	Voidage of fluidised bed
ϵ	Emmisivity (infrared radiation)
ϵ'	Dielecric constant (microwaves)
ϵ''	Loss factor (microwaves)
θ	Temperature
λ	Latent heat
λ	Wavelegth
μ	Viscosity
μ	Specific growth rate (fermentation)
Π	Osmatic pressure
π	Constant $= 3.142$
ρ	Density
Σ	Sum
σ	Standard deviation
σ	Electrical conductivity
σ	Stefan-Boltzmann constant (infrared radiation)
ω	Angular velocity